ETH zürich

D BSSE

Pseudo-temporal ordering and RNA velocity – Practical session

Zhisong He, PhD
Senior Scientist and Lecturer
Treutlein lab, D-BSSE, ETH Zurich
Basel, Switzerland
20.05.2026

Outlines

- **Introduction**
  - Setup your environment (Python)
  - The example data set

- **Analysis**
  - Brief overview on the data
  - Diffusion pseudotime
  - Coarse-grained trajectory analysis with PAGA
  - RNA velocity analysis with scVelo
  - Fate probability estimation with CellRank 2

Set up your work environment

Packages in need:
- Python
  - scanpy
  - scvelo
  - cellrank
  - (pyurd)

Set up local conda environment:

After installing conda (miniconda/miniforge/anaconda)…

```
> conda create -n env_embo_pseudotime python=3.11
> conda activate env_embo_pseudotime
> git clone https://github.com/GIMM-BioCode/2026_Pipette2Code.git
> cd 2026_Pipette2Code/02_Trajectories_ZHe
> pip install -r requirements.txt
```

Recommendation for Windows users

Install a Linux environment with WSL2

Use Google Colab

Register a Google account if you don't have one. Afterwards, login and visit the Colab website: https://colab.research.google.com/. Create a new notebook and open. In the notebook, input and execute the follows:

```
!git clone https://github.com/GIMM-BioCode/2026_Pipette2Code.git
!pip install -q -r 2026_Pipette2Code/02_Trajectories_ZHe/requirements-colab.txt git+https://github.com/zhisonghe/pyurd.git
```

More information can be seen here:
https://github.com/GIMM-BioCode/2026_Pipette2Code/tree/main/02_Trajectories_ZHe/README.md

Online vignette for the analysis

https://github.com/GIMM-BioCode/2026_Pipette2Code/tree/main/02_Trajectories_ZHe/

Files

main

Go to file T

- 01_Introduction_TGomes
- 02_Trajectories_ZHe
  - .gitignore
  - .python-version
  - README.md
  - Tutorial.ipynb
  - Tutorial_cleared.ipynb
  - pyproject.toml
  - requirements-colab.txt
  - requirements.txt
  - uv.lock
- 03_Integration_SKoplev
- 04_ImmuneReceptors_AMaceiras
- .gitignore
- README.md

2026_Pipette2Code / 02_Trajectories_ZHe / Add file

zhisonghe Rename notebook and update README 7e1cc18 · 23 minutes ago History

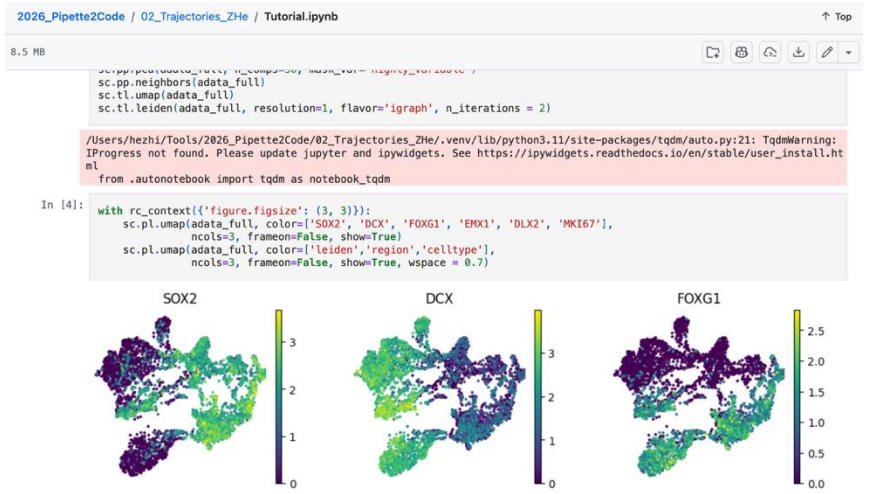
Name	Last commit message	Last commit date
..		
.gitignore	Add pyURD traject… …ironment setup	1 hour ago
.python-version	…d pyURD trajectory analysis section and environment setup	1 hour ago
	Rename notebook and update README	23 minutes ago
Tutorial.ipynb	Rename notebook and update README	23 minutes ago
Tutorial_cleared.ipynb	Rename notebook and update README	23 minutes ago
	Add pyURD trajectory analysis section and environment setup	1 hour ago
requirements-colab.txt	Add requirements-colab.txt for Google Colab compatibility	1 hour ago
requirements.txt	…y analysis section and environment setup	1 hour ago
uv.lock	Add pyURD trajectory analysis …t setup	1 hour ago

README.md

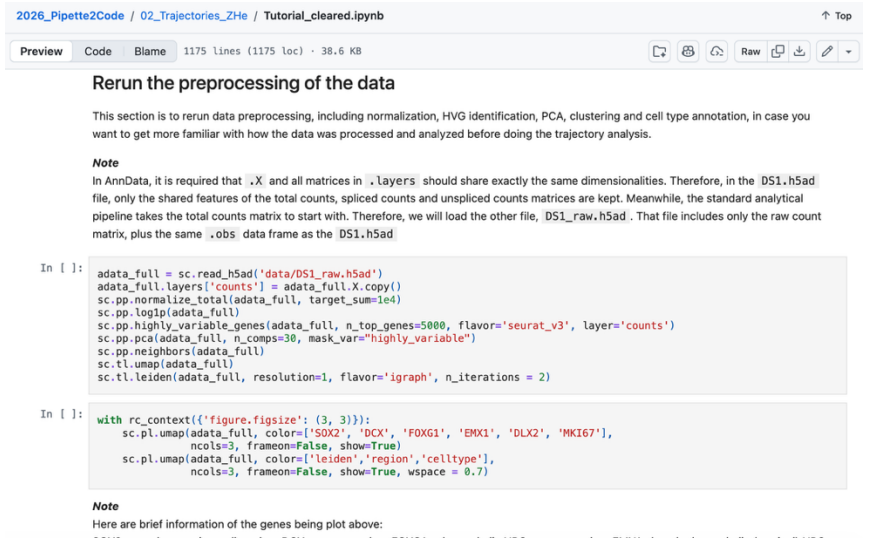
Trajectory and velocity analysis

This session is handled by Zhisong He

Both codes and outputs

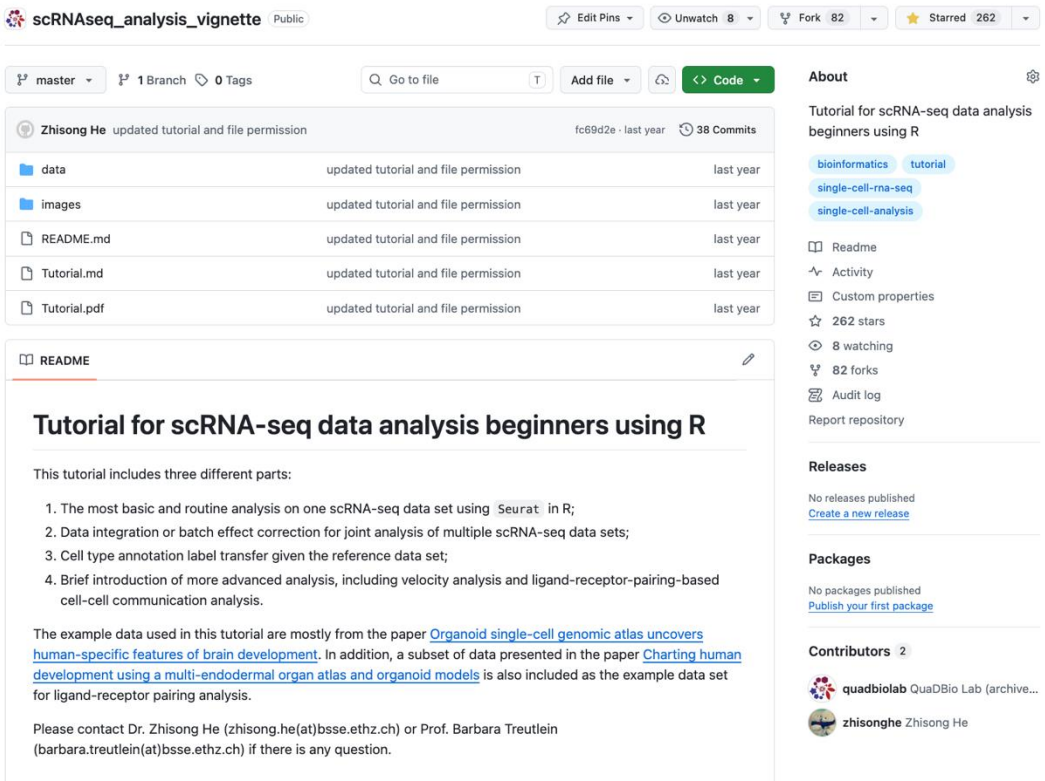


Code only

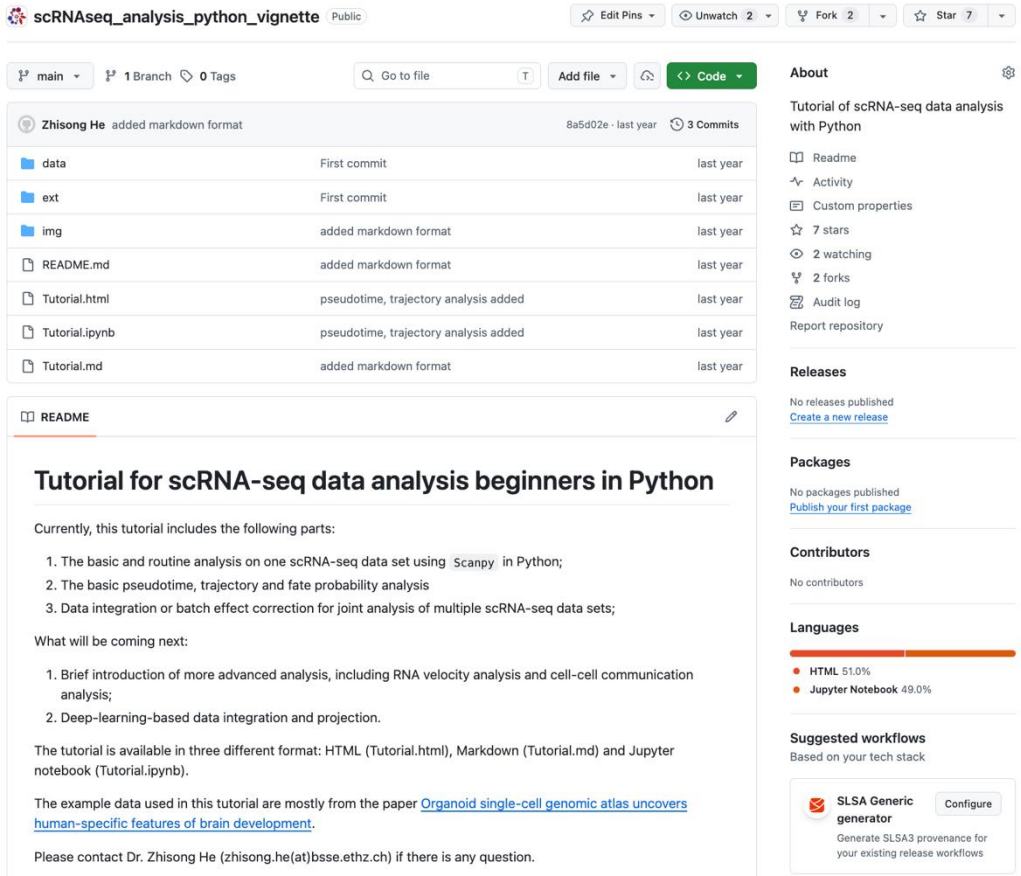


Alternative vignette for the analysis

R: https://github.com/quadbio/scRNAseq_analysis_vignette



Python: https://github.com/quadbio/scrnaseq_analysis_python_vignette/



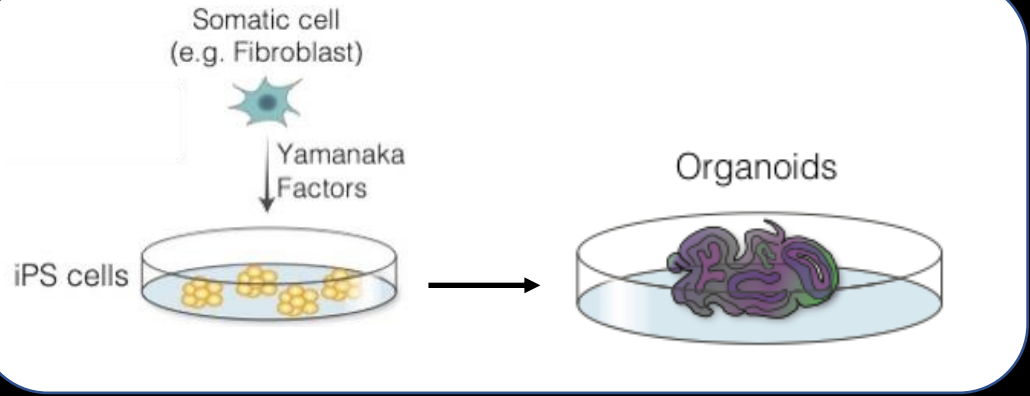
Example data set: scRNA-seq data of brain organoids

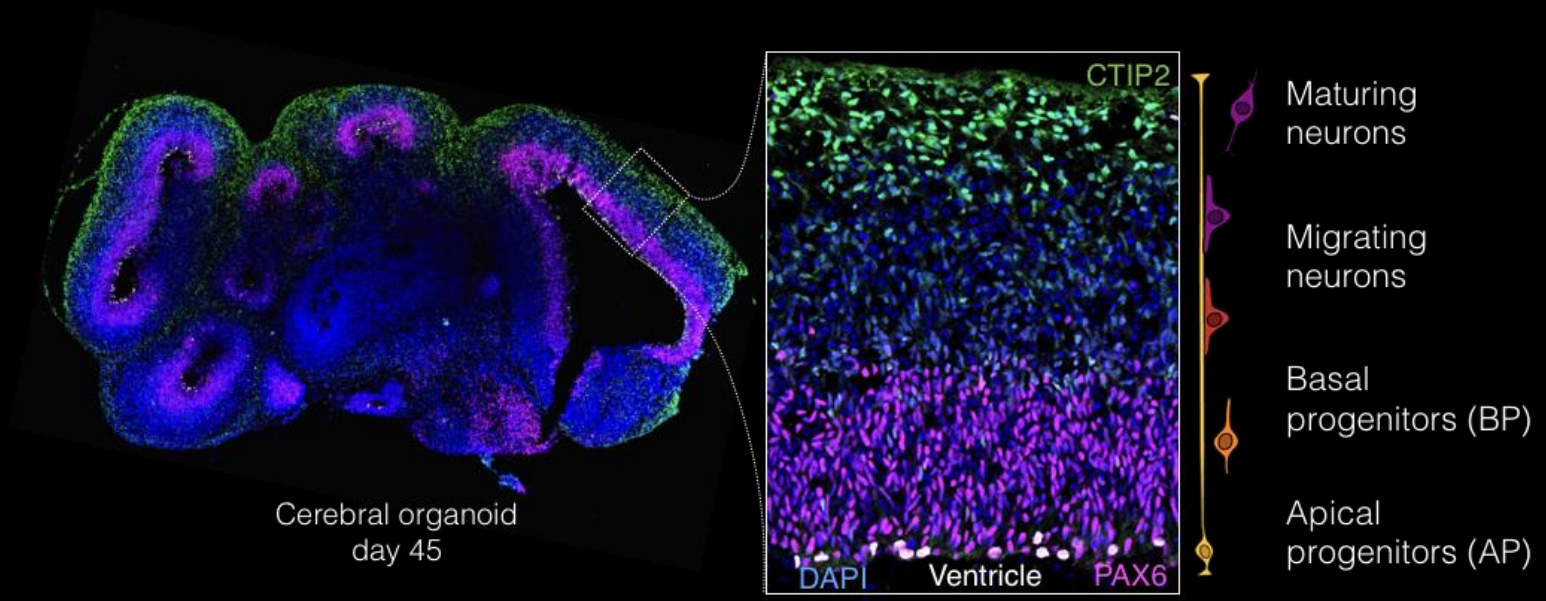
LETTER

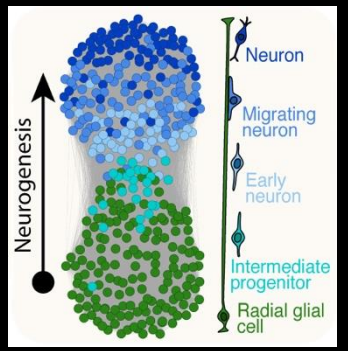
https://doi.org/10.1038/s41586-019-1654-9

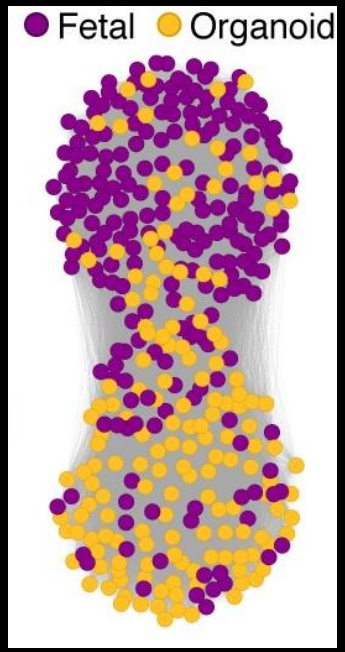
Organoid single-cell genomic atlas uncovers human-specific features of brain development

Sabina Kanton[1,7], Michael James Boyle[1,7], Zhisong He[1,2,7]*, Malgorzata Santel[1], Anne Weigert[1], Fátima Sanchís-Calleja[1,2], Patricia Guijarro[3], Leila Sidow[1], Jonas Simon Fleck[2], Dingding Han[3], Zhengzong Qian[3], Michael Heide[4], Wieland B. Huttner[4], Philipp Khaitovich[1,3,5], Svante Pääbo[1], Barbara Treutlein[1,2]* & J. Gray Camp[1,6]*



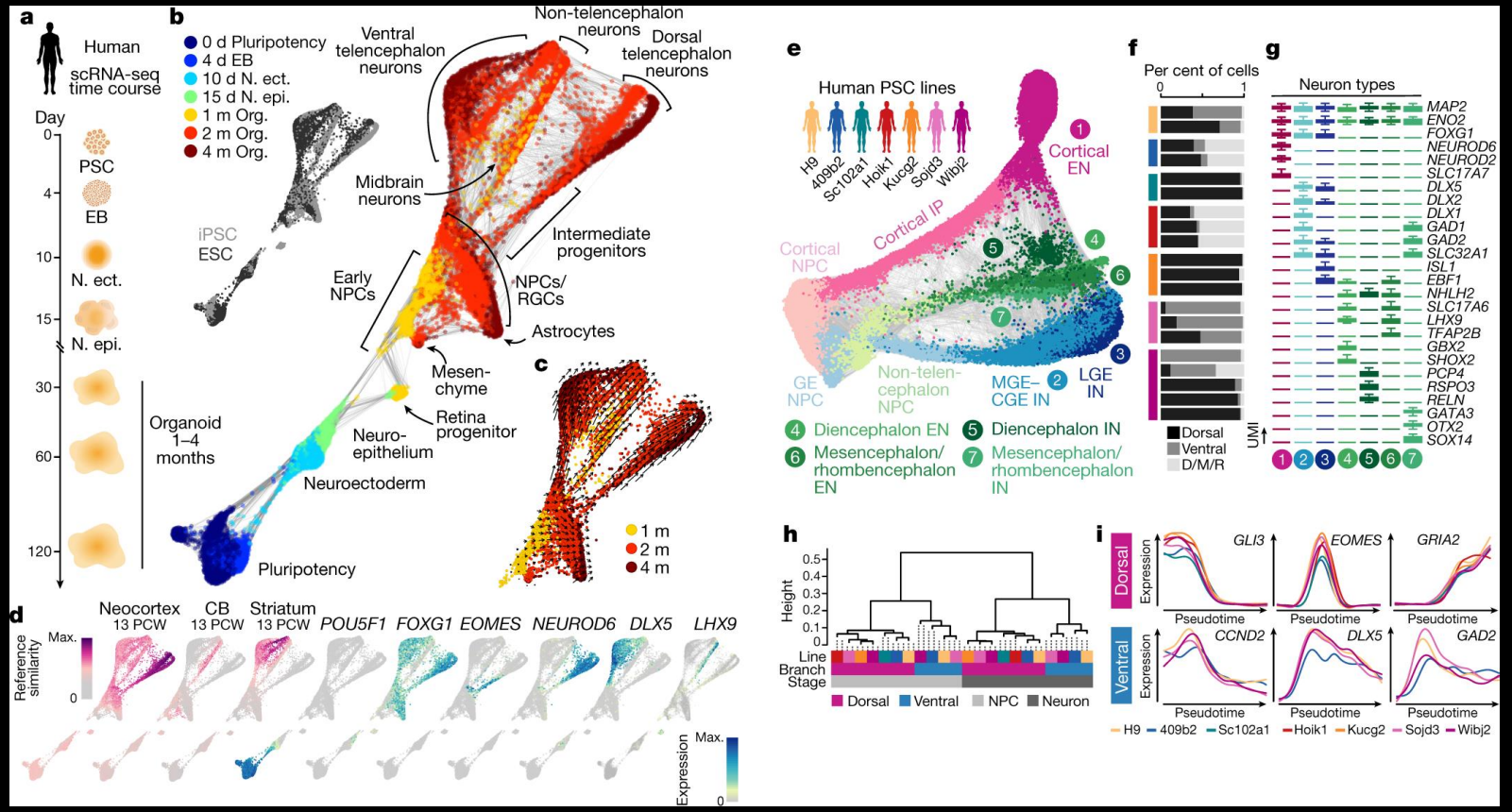


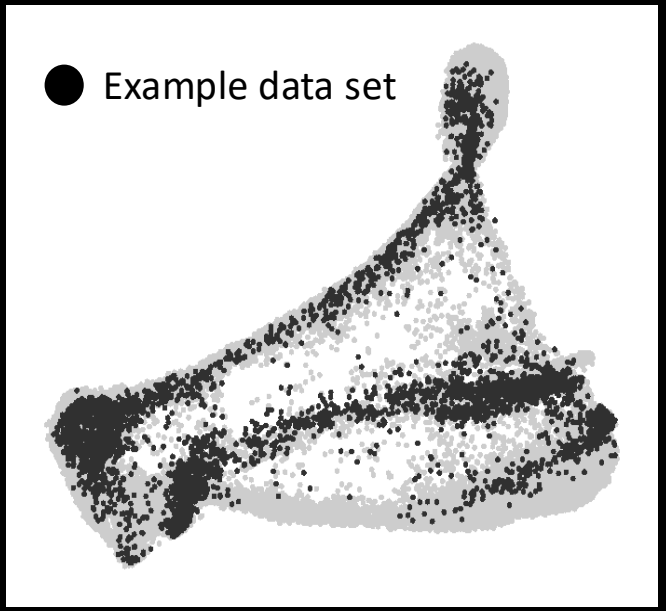




Camp et al. (2015) Human cerebral organoids recapitulate gene expression programs of fetal neocortex development. *PNAS*

Example data set: scRNA-seq data of brain organoids





Example data set

One 2-month-old organoid

- 4317 cells
- Seurat object
- Preprocessed and annotated
- Also include exonic/intronic count matrices as additional assays

Link to the data (H5AD files for the AnnData objects):

(1) With total, spliced and unspliced counts: https://polybox.ethz.ch/index.php/s/GCZzdxkMiTp5ZQH

(2) With only total counts but all features: https://polybox.ethz.ch/index.php/s/GeXSaepAk9fjQqk

Outlines

- **Introduction**
  - Setup your environment (Python)
  - The example data set

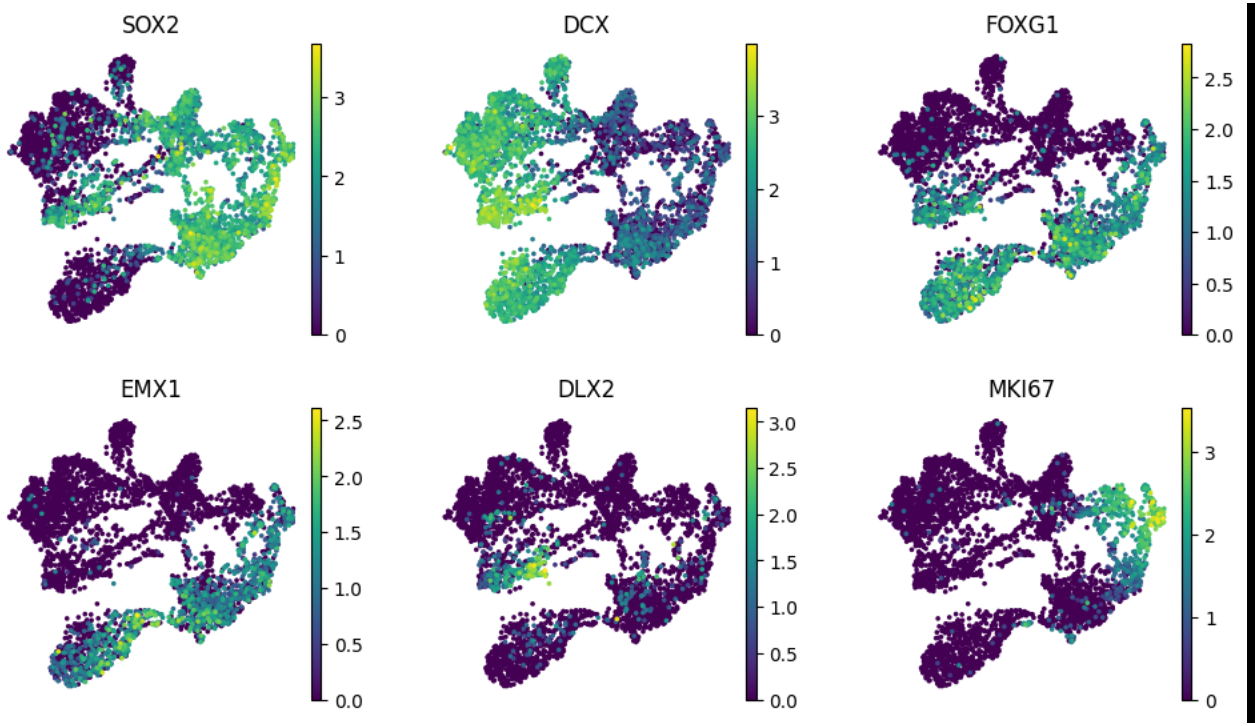
- **Analysis**
  - Brief overview on the data – 5 min
  - Diffusion pseudotime – 10 min
  - Coarse-grained trajectory analysis with PAGA – 10 min
  - RNA velocity analysis with scVelo – 15 min
  - Fate probability estimation with CellRank 2 – 15 min
  - Trajectory analysis with pyURD – 10 min

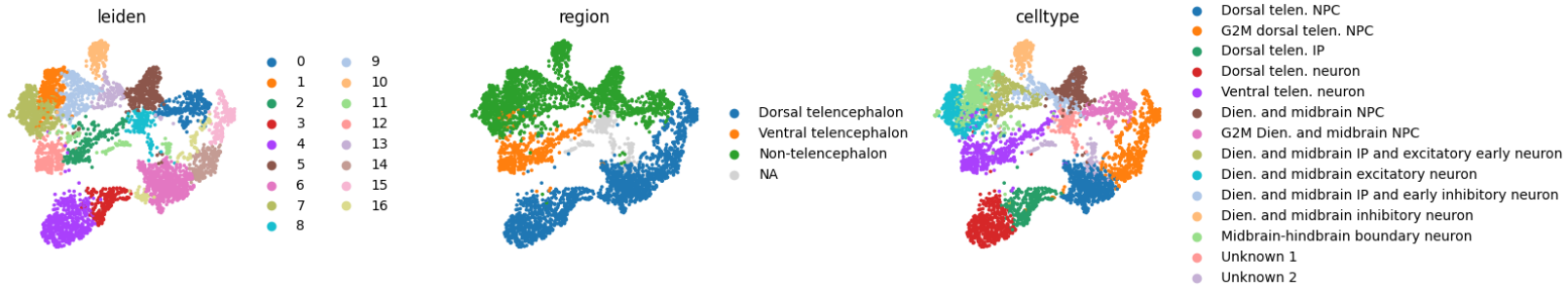
Standard analysis (with total count matrix)

```
import scanpy as sc
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from matplotlib.pyplot import rc_context

adata_full = sc.read_h5ad('data/DS1_raw.h5ad')
adata_full.layers['counts'] = adata_full.X.copy()
sc.pp.normalize_total(adata_full, target_sum=1e4)
sc.pp.log1p(adata_full)
sc.pp.highly_variable_genes(adata_full, n_top_genes=5000, flavor='seurat_v3', layer='counts')
sc.pp.pca(adata_full, n_comps=30, mask_var="highly_variable")
sc.pp.neighbors(adata_full)
sc.tl.umap(adata_full)
sc.tl.leiden(adata_full, resolution=1, flavor='igraph', n_iterations = 2)

with rc_context({'figure.figsize': (3, 3)}):
    sc.pl.umap(adata_full, color=['SOX2', 'DCX', 'FOXG1', 'EMX1', 'DLX2', 'MKI67'],
        ncols=3, frameon=False, show=True)
    sc.pl.umap(adata_full, color=['leiden','region','celltype'],
        ncols=3, frameon=False, show=True, wspace = 0.7)
```

Standard analysis (with total count matrix)

```
import scanpy as sc
import numpy as np
import pandas as pd
import matplotlib.pyplot
from matplotlib.pyplot im

adata_full = sc.read_h5ac
adata_full.layers['counts
sc.pp.normalize_total(ada
sc.pp.log1p(adata_full)
sc.pp.highly_variable_ger
sc.pp.pca(adata_full, n_c
sc.pp.neighbors(adata_ful
sc.tl.umap(adata_full)
sc.tl.leiden(adata_full,

with rc_context({'figure.
    sc.pl.umap(adata_full
        ncols=3, frameon
    sc.pl.umap(
        ncols=3
```





Diffusion map and DPT

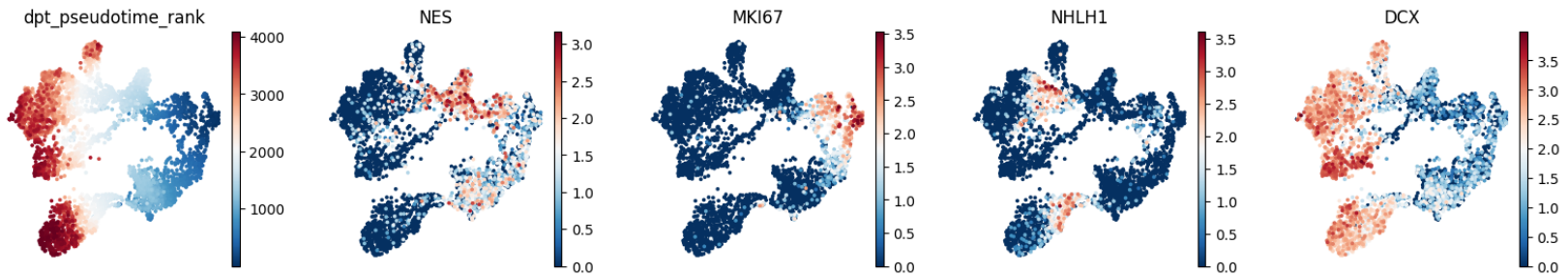
```
# Load the main object
adata = sc.read_h5ad('data/DS1.h5ad')

# Copy the preprocessed results to the main object
adata.obsm = adata_full[adata.obs_names, :].obsm.copy()
adata.obsp = adata_full[adata.obs_names, :].obsp.copy()
adata.uns = adata_full[adata.obs_names, :].uns.copy()

# Run Diffusion Map
sc.pp.neighbors(adata, n_neighbors=50, n_pcs=30, use_rep='X_pca')
sc.tl.diffmap(adata, n_comps=20)
```

```
# Infer the root
idx_subset = np.where(
    np.isin(
        adata.obs['celltype'],
        [
            'Dorsal telen. NPC',
            'G2M dorsal telen. NPC',
            'Dien. and midbrain NPC',
            'G2M Dien. and midbrain NPC'
        ]
))[0]
idxs_rand_root = np.apply_along_axis(
    lambda x: random_root(adata, idx_subset=idx_subset),
    1, np.array(range(1000))[:, None]
)
adata.uns['iroot'] = np.argmax(np.bincount(idxs_rand_root))
```

```
# Method to infer the root
import random
def random_root(
    adata, seed=None, neighbors_key=None, idx_subset=None
):
    if seed is not None:
        random.seed(seed)
    iroot_bak = None
    if 'iroot' in adata.uns.keys():
        iroot_bak = adata.uns['iroot'].copy()
    dpt_bak = None
    if 'dpt_pseudotime' in adata.obs.columns:
        dpt_bak = adata.obs['dpt_pseudotime'].copy()

    idx = np.random.choice(list(range(adata.shape[0])))
    adata.uns['iroot'] = idx
    sc.tl.dpt(adata, neighbors_key=neighbors_key)
    dpt = adata.obs['dpt_pseudotime']
    if idx_subset is not None:
        dpt = dpt.iloc[idx_subset]
    idx_max_dpt = np.argmax(dpt)
    if idx_subset is not None:
        idx_max_dpt = idx_subset[idx_max_dpt]

    del adata.uns['iroot']
    del adata.obs['dpt_pseudotime']
    if iroot_bak is not None:
        adata.uns['iroot'] = iroot_bak.copy()
    if dpt_bak is not None:
        adata.obs['dpt_pseudotime'] = dpt_bak.copy()

    return idx_max_dpt
```

Diffusion map and DPT

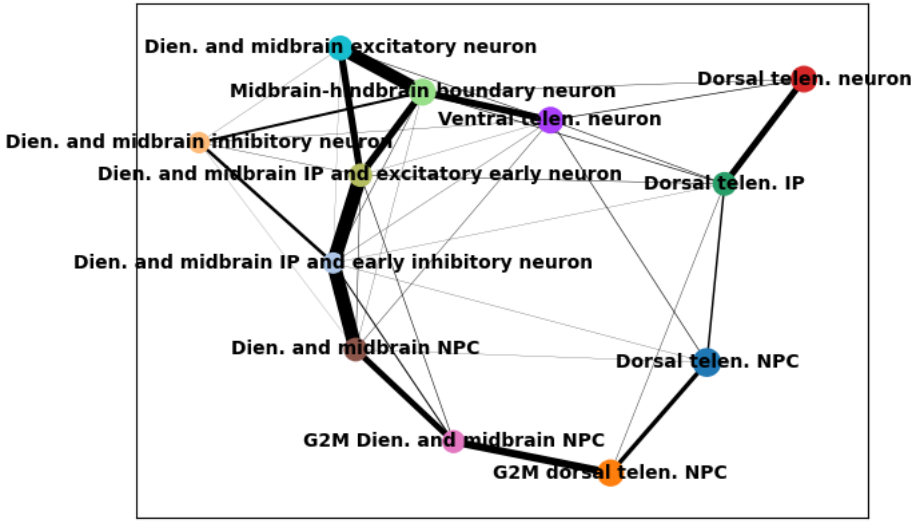
```
# Run DPT
sc.tl.dpt(adata, n_dcs=20)

# Rank DPT
from scipy.stats import rankdata
adata.obs['dpt_pseudotime_rank'] = rankdata(adata.obs['dpt_pseudotime'])

# Visualization of DPT on UMAP
with rc_context({'figure.figsize': (3, 3)}):
    sc.pl.umap(
        adata,
        color=['dpt_pseudotime_rank', 'NES', 'MKI67', 'NHLH1', 'DCX'],
        color_map='RdBu_r',
        ncols=5, frameon=False
    )
```



Coarse-grained trajectory analysis with PAGA

```
sc.pp.neighbors(adata, n_neighbors=50, n_pcs=20, use_rep='X_diffmap')
adata.obs['celltype'] = adata.obs['celltype'].cat.remove_unused_categories()
sc.tl.paga(adata, groups='celltype')
sc.pl.paga(adata)
```

Perform PAGA and visualize estimated cell type connectivities




```
from scipy import sparse


connected = adata.uns['paga']['connectivities'] > 0.1
connected = (connected + connected.T) > 0
idx_row, idx_col, dat = sparse.find(connected)
idx = (idx_row >= idx_col)
connected_celltypes = pd.DataFrame(
    {
        'node1': adata.obs['celltype'].cat.categories[idx_row[idx]],
        'node2': adata.obs['celltype'].cat.categories[idx_col[idx]]
    }
)
connected_celltypes
```

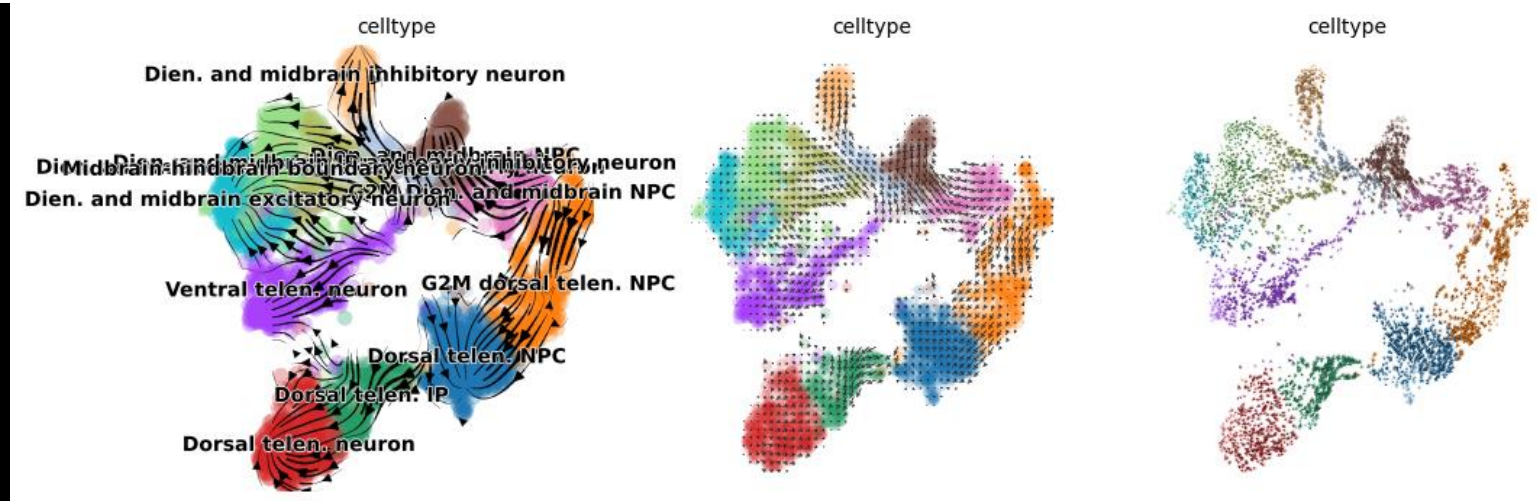
Check connected cell types

RNA velocity analysis with scVelo

```
import scvelo as scv

# Filter features
adata.raw = adata
scv.pp.filter_and_normalize(
    adata, min_shared_counts=20,
    min_shared_cells=20
)

# Estimate velocity
sc.pp.neighbors(
    adata, use_rep='X_pca', n_neighbors=50
)
scv.pp.moments(adata, n_neighbors=50)
scv.tl.velocity(adata, mode='stochastic')
scv.tl.velocity_graph(adata)

# Visualization
fig, ax = plt.subplots(1, 3, figsize=(12, 4))
scv.pl.velocity_embedding_stream(
    adata, ax=ax[0], basis='umap', color='celltype', frameon=False, show=False
)
scv.pl.velocity_embedding_grid(
    adata, ax=ax[1], basis='umap', color='celltype', frameon=False, arrow_size=2, arrow_length=2, show=False
)
scv.pl.velocity_embedding(
    adata, ax=ax[2], basis='umap', color='celltype', frameon=False, arrow_size=2, arrow_length=2, show=False
)
plt.show()
```

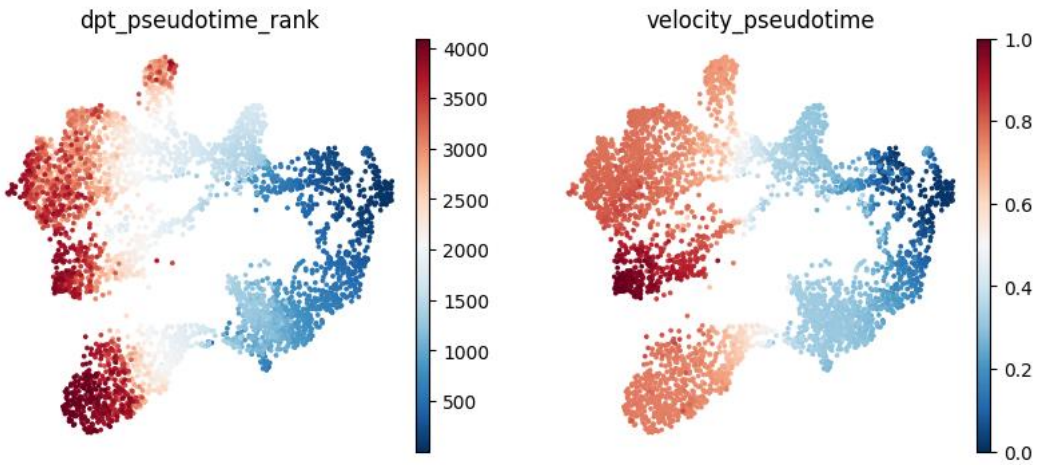
RNA velocity analysis with scVelo

```
import scvelo as scv

# Filter features
adata.raw = adata
scv.pp.filter_and_normalize(
    adata, min_shared_counts=20,
    min_shared_cells=20
)

# Estimate velocity
sc.pp.neighbors(
    adata, use_rep='X_pca', n_neighbors=50
)
scv.pp.moments(adata, n_neighbors=50)
scv.tl.velocity(adata, mode='stochastic')
scv.tl.velocity_graph(adata)

# Visualization
fig, ax = plt.subplots(1, 3, figsize=(12, 4))
scv.pl.velocity_embedding_stream(
    adata, ax=ax[0], basis='umap', color='celltype', frameon=False, show=False
)
scv.pl.velocity_embedding_grid(
    adata, ax=ax[1], basis='umap', color='celltype', frameon=False, arrow_size=2, arrow_length=2, show=False
)
scv.pl.velocity_embedding(
    adata, ax=ax[2], basis='umap', color='celltype', frameon=False, arrow_size=2, arrow_length=2, show=False
)
plt.show()
```



RNA velocity analysis with scVelo

```
# Estimate velocity pseudotime
scv.tl.velocity_pseudotime(adata)

# Visualize both the DPT and velocity pseudotime
with rc_context({'figure.figsize': (4, 4)}):
    sc.pl.umap(adata, color=['dpt_pseudotime_rank', 'velocity_pseudotime'],
    cmap='RdBu_r', frameon=False, ncols=2)
```



Fate probability estimation with CellRank2

```
import cellrank as cr

pk = cr.kernels.PseudotimeKernel(adata, time_key='dpt_pseudotime').compute_transition_matrix()
ck = cr.kernels.ConnectivityKernel(adata).compute_transition_matrix()
vk = cr.kernels.VelocityKernel(adata).compute_transition_matrix()
combined_kernel = 0.5 * vk + 0.3 * pk + 0.2 * ck
```

Generate hybrid kernels summarizing velocity, transcriptomic similarity (connectivity) and pseudotime

```
g = cr.estimators.GPCCA(combined_kernel)
g.fit(n_states=15, cluster_key='celltype'
g.predict_terminal_states(method='top_n', n_states=10)
g.plot_macrostates(which='terminal')
## Google Colab users may fail at this step. It won't affect the following steps.
```

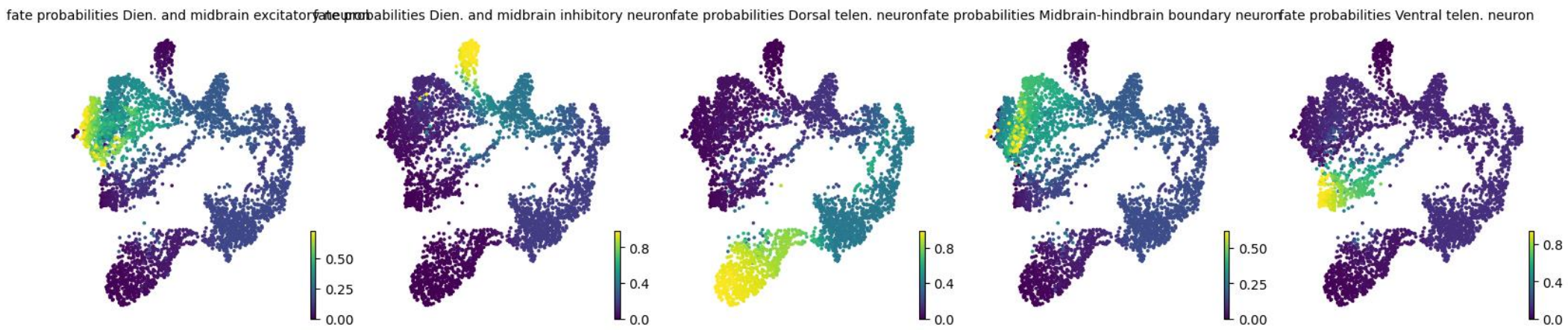
Predict terminal states

```
g = cr.estimators.GPCCA(combined_kernel)
neuron_types = [x for x in adata.obs['celltype'].cat.categories if x.endswith('neuron') and 'early' not in x]
terminal_states = [adata[adata.obs['celltype'] == x,
:].obs['dpt_pseudotime'].sort_values(ascending=False)[:30].index
for x in neuron_types]
terminal_states = dict(zip(neuron_types, terminal_states))
g.set_terminal_states(terminal_states)
g.plot_macrostates(which='terminal')
```

Manually assign terminal states (each neuron type with the highest diffusion pseudotime)

Fate probability estimation with CellRank2

```
g.compute_fate_probabilities()
with rc_context({'figure.figsize': (4, 4)}):
    g.plot_fate_probabilities(legend_loc='right', basis='X_umap', same_plot=False)
```

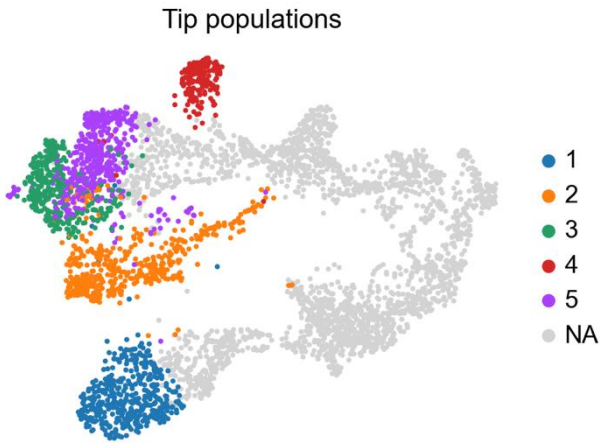



```
# Save the results
prob = pd.DataFrame(g.fate_probabilities).set_index(g.terminal_states.index).set_axis(g.terminal_states.cat.categories, axis=1)
adata.obs = pd.concat([adata.obs, prob.set_axis(['CR_' + x for x in prob.columns], axis=1)], axis=1)
for obsm in adata.obsm.keys():
    if type(adata.obsm[obsm]).__name__ in ['Lineage','LineageView']:
        adata.uns[str(obsm)+'_lineage'] = {
            'colors': adata.obsm[obsm].colors,
            'names': adata.obsm[obsm].names
        }
        adata.obsm[obsm] = np.array(adata.obsm[obsm])
adata.write_h5ad('data/DS1_processed.h5ad')
```

(Optional) Trajactory analysis with pyURD

```
# Load package and data
import pyurd
adata_urd = sc.read_h5ad("data/DS1_processed.h5ad")
```

```
# Root cells: all NPC populations
ROOT_TYPES = [
    "Dorsal telen. NPC",
    "G2M dorsal telen. NPC",
    "Dien. and midbrain NPC",
    "G2M Dien. and midbrain NPC",
]
root_cells = adata_urd.obs_names[adata_urd.obs["celltype"].isin(ROOT_TYPES)].tolist()

# Tip populations: five terminal neuronal lineages
TIP_MAP = {
    "1": "Dorsal telen. neuron",
    "2": "Ventral telen. neuron",
    "3": "Dien. and midbrain excitatory neuron",
    "4": "Dien. and midbrain inhibitory neuron",
    "5": "Midbrain-hindbrain boundary neuron",
}
adata_urd.obs["tip"] = np.nan
for tip_id, ct in TIP_MAP.items():
    mask = adata_urd.obs["celltype"] == ct
    adata_urd.obs.loc[mask, "tip"] = tip_id

# Visualize the tip populations
sc.pl.umap(adata_urd, color="tip", title="Tip populations")
```

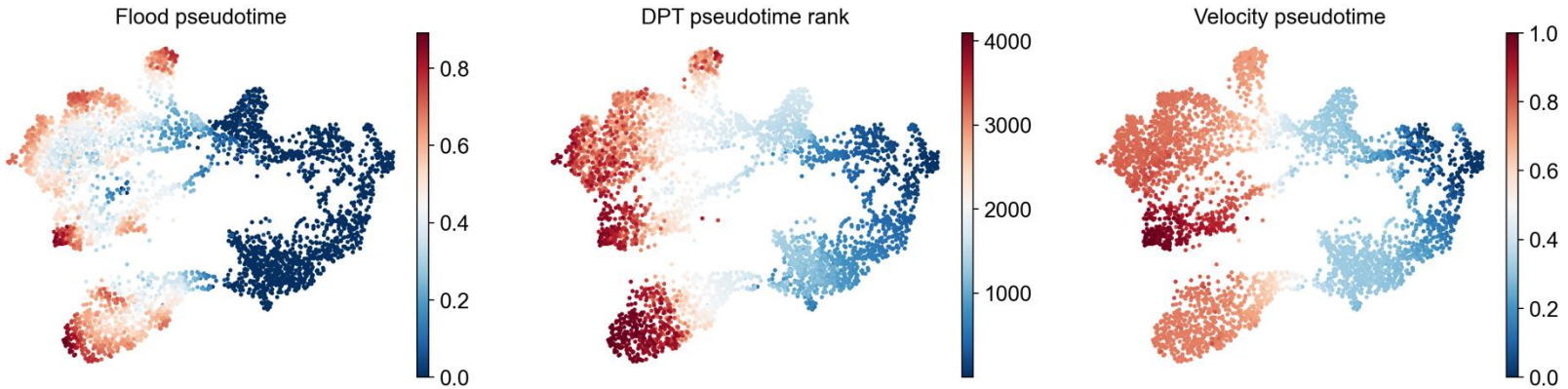


(Optional) Trajactory analysis with pyURD

```
# Diffusion map based transition matrix denoising
transition_key = pyurd.prepare_connectivities(adata_urd, n_neighbors=50, n_dcs=30)

# Flood simulations
floods = pyurd.flood_pseudotime(
    adata_urd,
    root_cells=root_cells,
    n=50, minimum_cells_flooded=2, transition_key=transition_key, verbose=False, n_jobs=-1,
)

# Summarize flood simulations into pseudotime
pyurd.flood_pseudotime_process(
    adata_urd,
    floods=floods,
    floods_name="pseudotime", max_frac_na=0.4, stability_div=10,
)
```

(Optional) Trajactory analysis with pyURD

```
# Visualize and compare different pseudotimes
sc.pl.umap(
    adata_urd,
    color=["pseudotime",'dpt_pseudotime_rank', 'velocity_pseudotime'],
    color_map="RdBu_r", ncols=3, frameon=False, title=["Flood pseudotime", "DPT pseudotime rank", "Velocity pseudotime"]
)
corr_df = adata_urd.obs[["pseudotime", "dpt_pseudotime_rank", "velocity_pseudotime"]].corr(method='spearman')
print(corr_df.to_string())
```



(Optional) Trajactory analysis with pyURD

```
# Fit logistic parameters:
logistic_params = pyurd.pseudotime_determine_logistic(
    adata_urd, pseudotime_key="pseudotime", optimal_cells_forward=20, max_cells_back=40, do_plot=True, verbose=True,
)

# Build the biased transition matrix
biased_tm = pyurd.pseudotime_weight_transition_matrix(
    adata_urd, pseudotime_key="pseudotime", logistic_params=logistic_params, transition_key=transition_key, chunk_size=500,
)
```

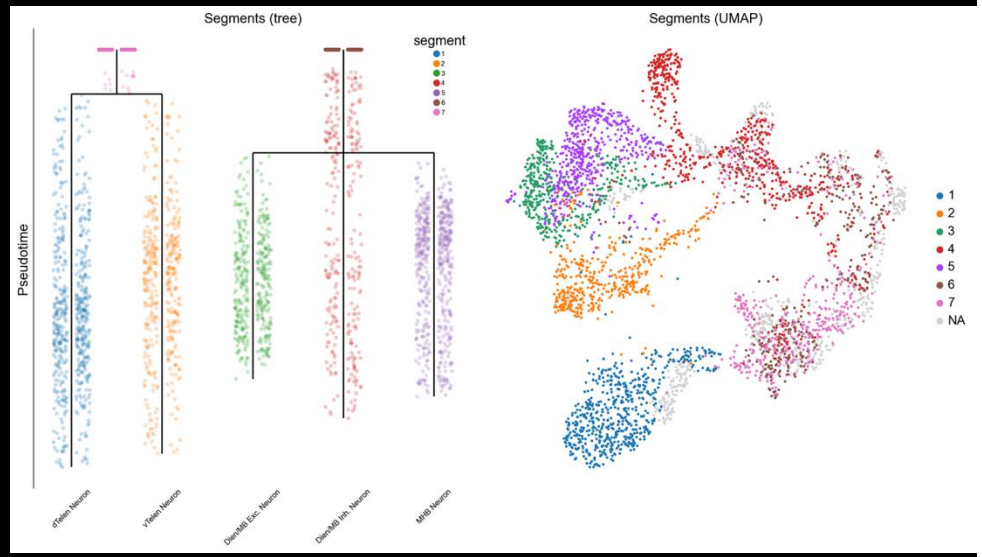
```
# Register tip cells
pyurd.load_tip_cells(adata_urd, tips_key="tip")

# Simulate 10 000 random walks per tip
walks = pyurd.simulate_random_walks_from_tips(
    adata_urd, tip_group_key="tip", root_cells=root_cells, transition_matrix=biased_tm, n_per_tip=10_000, root_visits=3,
    max_steps=5000, verbose=True,
)

# Convert walks to per-cell visitation frequencies stored in adata_urd.obs
pyurd.process_random_walks_from_tips(adata_urd, walks_dict=walks, verbose=True)
```

(Optional) Trajactory analysis with pyURD

```
# Join the five tip trajectories by finding the pseudotime at which their visitation patterns diverge
pyurd.build_tree(
    adata_urd, pseudotime_key="pseudotime", tips_use=list(TIP_MAP.keys()), # ["1", "2", "3", "4", "5"]
    divergence_method="preference", # uses Hartigan dip test
    weighted_fusion=True, cells_per_pseudotime_bin=80, bins_per_pseudotime_window=5, minimum_visits=10,
    visit_threshold=0.7, p_thresh=0.01, dendro_node_size=100, verbose=True,
)

# Assign human-readable names to terminal segments
pyurd.name_segments(
    adata_urd, segments=["1", "2", "3", "4", "5"],
    segment_names=[
        "dTelen Neuron",
        "vTelen Neuron",
        "Dien/MB Exc. Neuron",
        "Dien/MB Inh. Neuron",
        "MHB Neuron",
    ], short_names=["dTN", "vTN", "DiMB-E", "DiMB-I", "MHB"],
)

# Visualize the tree and segments on the UMAP
fig, axes = plt.subplots(1, 2, figsize=(14, 8))
pyurd.plot_tree(adata_urd, label="segment", title="Segments (tree)", ax=axes[0], legend=True, cell_size=4)
sc.pl.umap(adata_urd, color="segment", title="Segments (UMAP)", ax=axes[1], show=False)
axes[1].set_aspect("auto")
plt.tight_layout()
plt.show()
```

(Optional) Trajectory analysis with pyURD

```
# Join the five tip trajectories by finding the pseudotime at which their visitation patterns diverge
pyurd.build_tree(
    adata_urd, pseudotime_key="pseudotime", tips_use=list(TIP_MAP.keys()), # ["1", "2", "3", "4", "5"]
    divergence_method="preference", # uses Hartigan dip test
    weighted_fusion=True, cells_per_pseudotime_bin=80, bins_per_pseudotime_window=5, minimum_visits=10,
    visit_threshold=0.7, p_thresh=0.01, dendro_node_size=100, verbose=True,
)

# Assign human-readable names to terminal segments
pyurd.name_segments(
    adata_urd, segments=["1", "2", "3", "4", "5"],
    segment_names=[
        "dTelen Neuron",
        "vTelen Neuron",
        "Dien/MB Exc. Neuron",
        "Dien/MB Inh. Neuron",
        "MHB Neuron",
    ], short_names=["dTN", "vTN", "DiMB-E", "DiMB-I", "MHB"],
)

# Visualize the tree and segments on the UMAP
fig, axes = plt.subplots(1, 2, figsize=(14, 8))
pyurd.plot_tree(adata_urd, label="segment", title="Segments (tree)", ax=axes[0], legend=True, cell_size=4)
sc.pl.umap(adata_urd, color="segment", title="Segments (UMAP)", ax=axes[1], show=False)
axes[1].set_aspect("auto")
plt.tight_layout()
plt.show()
```



Questions?